Signal, Image and Video Processing (2026) 20:138
https://doi.org/10.1007/s11760-026-05166-z

ORIGINAL PAPER

# HF-YOLO: A Model for High Frequency Signal Detection and Recognition in Wideband

**Libin Gu[1] · Tianxue Chen[2] · Kai Liu[1]**

Received: 29 August 2025 / Revised: 9 October 2025 / Accepted: 29 January 2026
© The Author(s), under exclusive licence to Springer-Verlag London Ltd., part of Springer Nature 2026

**Abstract**
Wideband high frequency (HF) signal detection is crucial for radio intelligence operations. However, multiple signals coexisting within a wideband spectrum, combined with dispersive and time-varying HF channels, complicates signal detection. This paper proposes a novel method called High Frequency-You Only Look Once (HF-YOLO) to address the above problems. Specifically, HF-YOLO processes wideband time-frequency image (WTFI) as input. An Inception feature extraction (IFE) module, based on the Inception architecture and group convolution, is proposed to improve the modeling of strip-distributed signals in WTFI while reducing computational complexity. Additionally, a time-frequency attention (TFA) module is introduced and improved considering the narrow bandwidth and strong temporal correlation of HF signals. The TFA effectively suppresses out-of-band interference and improves the detection of weak signals. To evaluate performance, a dataset called wideband time-frequency image dataset (WTFID) is constructed comprising a variety of HF signals covering diverse communication scenarios. Experimental results demonstrate that HF-YOLO achieves high detection accuracy and real-time performance even in signal-to-noise ratio (SNR) below 0 dB, confirming the effectiveness of HF-YOLO under complex channel conditions.

**Keywords** Signal detection · Wideband spectrogram · High frequency · YOLO

## 1 Introduction

High frequency (HF) communication [1, 2] is widely used in military and civil fields due to its low cost and flexible deployment. Wideband HF signal detection-including existence detection, parameter estimation and type recognition-is an essential step in HF communications. However, multiple target signals with overlapping durations and different frequencies in wideband make parameter estimation difficult. Furthermore, the HF channel [3] is vulnerable to multipath effects and Doppler shift, which result in low signal-to-noise ratio (SNR) and further hinder reliable detection.

✉ Kai Liu
liukai@shu.edu.cn

Libin Gu
galibing@shu.edu.cn

Tianxue Chen
tianxuec@shu.edu.cn

1 School of Communication and Information Engineering, Shanghai University, Shanghai, China

2 Research & Development Group, iFlytek, Hefei, China

Traditional HF signal detection methods divide the wideband spectrum into multiple narrow subbands and rely on predefined template libraries [4–6]. These methods struggle to adapt to channel variations in low SNR environments and fail to meet the demands of real-time performance. To improve detection efficiency and accuracy, object detection algorithms have been increasingly applied to communication signal detection tasks [7]. In particular, You Only Look Once (YOLO) algorithms [8] enhance inference speed while maintaining detection performance. Some researchers have applied YOLO algorithms to signal detection, though their performance still needs improvement in low SNR conditions [9, 10].

To address these issues, this paper proposes the High Frequency-You Only Look Once (HF-YOLO) algorithm based on time-frequency characteristics of signals. We use wideband time-frequency images (WTFIs) as inputs. Specifically, we provide an Inception feature extraction (IFE) module and a time-frequency attention (TFA) mechanism to capture time-frequency features of signals effectively. The contributions of this paper are as follows:

Published online: 09 March 2026

1. We propose a lightweight HF signal detection algorithm called HF-YOLO. Considering the strip-like energy distribution of HF signals in WTFIs, we design the IFE module to strengthen the ability to distinguish between signal and noise. IFE uses group convolutions to reduce redundant channel computations. Additionally, we introduce the TFA module, which performs joint time-frequency modeling to capture energy distribution differences of various signals.
2. We construct a dataset named wideband time-frequency image dataset (WTFID), which encompasses various modulation schemes and application scenarios. Experiments conducted on this dataset demonstrate that HF-YOLO achieves high-precision detection of target signals, exhibiting excellent performance in low SNR and complex environments.

The remainder of this paper is organized as follows. In Section 2, some related works of wideband HF signal detection are introduced. Section 3 analyses the problem definition of wideband HF signal detection. Section 4 describes the proposed method in detail. Section 5 introduces the dataset and presents the experimental performance. Finally, Section 6 concludes this paper.

## 2 Related work

This section respectively summarizes the research progress of wideband signal detection and HF signal recognition in Subsection 2.1 and Subsection 2.2.

### 2.1 Wideband signal detection

Due to high detection accuracy and fast processing speed, object detection algorithms have been widely adopted for wideband signal detection. O'Shea et al. [9] first introduced the YOLO algorithm to signal detection. By training two-dimensional time-frequency waterfall images with bounding box annotations, they estimated signal frequency position as well as start and end time. Zha et al. [11] designed a detection model based on Single Shot MultiBox Detector (SSD) and utilized eye diagrams and vector maps to identify various signal modulation schemes above 0 dB. Prasad et al. [12] optimized the Faster-RCNN framework, reducing computational complexity while maintaining detection accuracy. Li et al. [10] optimized the scale and aspect ratio of anchor boxes based on the slender structural characteristics of communication signals in spectrum diagrams and improved detection accuracy using YOLOv3.

Power spectral distribution features are also widely used for detection in addition to time-frequency characteristics. Huang et al. [13] treated wideband power spectrum as a one-dimensional image and proposed a model based on Convolutional Neural Network (CNN) to directly estimate center frequency and bandwidth. Lin et al. [14] employed Welch periodograms to reduce the variance of power spectral density estimation and introduced a pyramid pooling module to integrate multi-scale contextual information, further improving the ability to distinguish spectral boundaries.

### 2.2 HF signal recognition

Traditional HF signal recognition methods [4, 5] mainly rely on template matching strategies to achieve target signal detection. Bezruk et al. [6] combined frequency-domain features with statistical learning theory and proposed a recognition scheme more suitable for low SNR and unknown signal scenarios. However, traditional methods depend on manually designed feature templates with high computational complexity, which limits their adaptability to the increasingly complex HF environment.

Deep learning-based methods identify signals by extracting features in time-frequency domain and designing specific neural network models. Xu et al. [15] introduced CNN to HF signal recognition tasks and verified the effectiveness of their algorithm on real-world signals. Yuan et al. [16] proposed a multi-stage recognition approach using a multi-channel signal acquisition module to screen candidate signals from wideband spectrum. Then they constructed a CNN network for feature extraction and pattern classification of candidate signals, ultimately achieving precise recognition of Morse code signals.

## 3 Problem analysis

Wideband HF signal detection, as the first step in blind processing of received signals, provides essential information for subsequent tasks. The key tasks are as follows:

1) Signal existence detection: Analyze the received wideband signal $r(n)$ to determine if the target signal is present. This can be modeled as a binary hypothesis testing problem:

$$\begin{aligned} &\mathrm{H}_0 : r(n) = g(n) \\ &\mathrm{H}_1 : r(n) = \sum_{i=1}^{N} x_i(n) + g(n) \end{aligned} \tag{1}$$

where, $N$ is the number of target signals in $r(n)$, $x_i(n)$ represents the $i$-th target signal, $g(n)$ stands for additive noise, $\mathrm{H}_0$ indicates that the received signal contains only noise, and $\mathrm{H}_1$ indicates that the target signal exists.

2) Signal parameter estimation and type recognition: In practice, the received signal often contains multiple target carrier signals at different frequencies with uncertain occur-

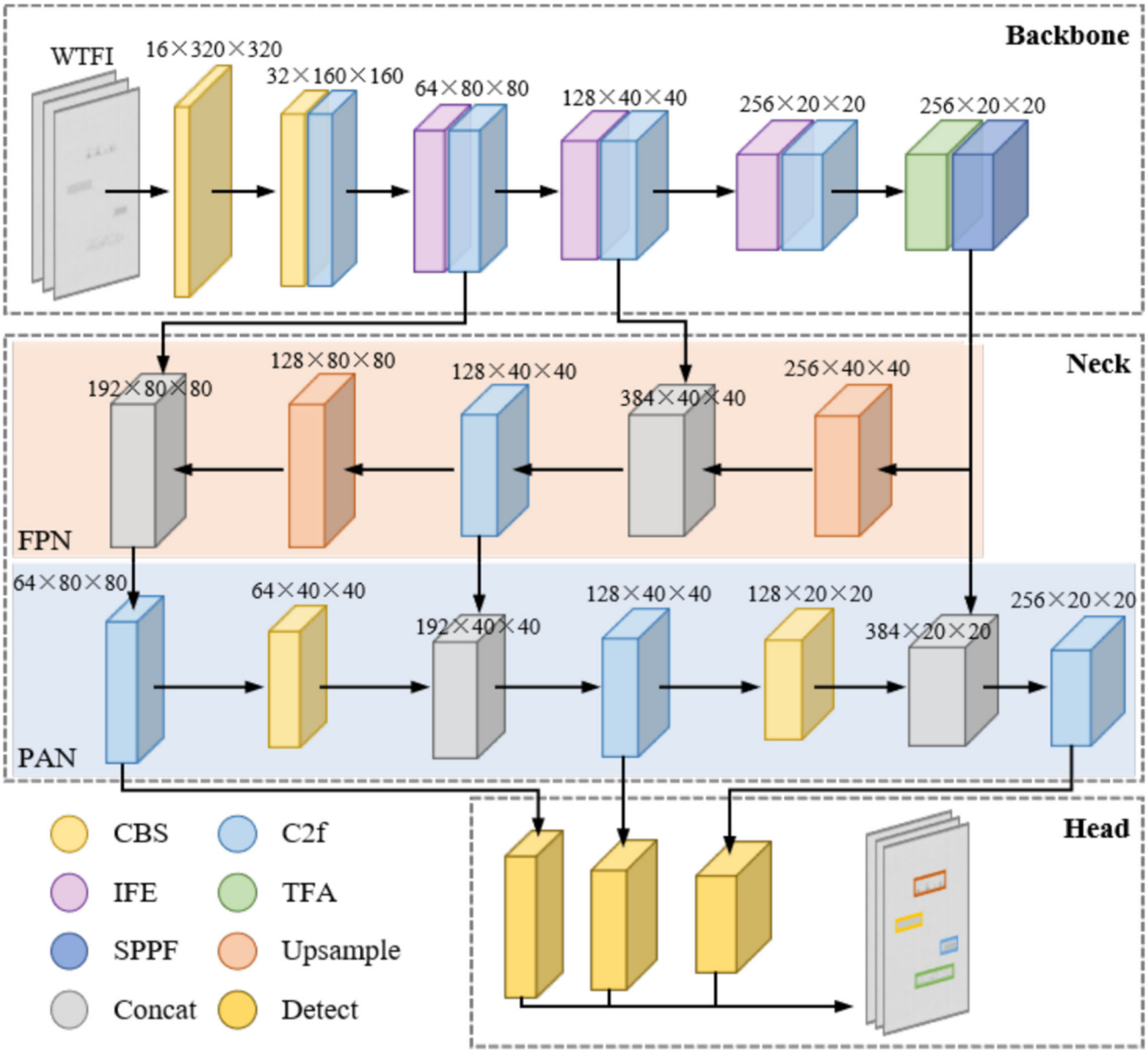



**Fig. 1** Overview of the proposed HF-YOLO, where $a \times b \times c$ denotes the dimensions of each layer

rence times, which increases detection difficulty. Following the signal existence detection, key parameters-including carrier center frequency, signal bandwidth and start and end time-are estimated and the signal type is subsequently recognized. The signal type is closely related to modulation methods and channel coding schemes, providing a foundation for subsequent in-depth analysis.

## 4 Proposed HF-YOLO method

This section presents a method called HF-YOLO for HF signal detection. Fig. 1 shows the overview of the proposed HF-YOLO, including three core parts: the backbone for feature extraction, the neck for feature fusion and the head for target prediction. To extract the time-frequency features of HF signals more effectively, HF-YOLO improves YOLOv8n [17] by replacing the Conv-BN-SiLU (CBS) module with the IFE module for multi-scale feature extraction and inserting the TFA module before the Spatial Pyramid Pooling Fast (SPPF) module to enhance joint time-frequency modeling in backbone.

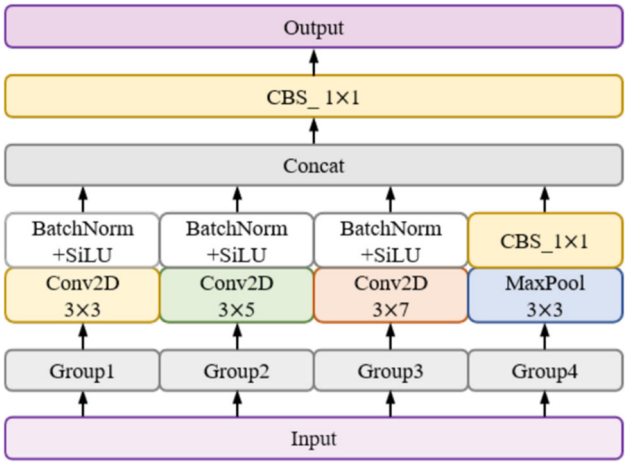



**Fig. 2** Structure of IFE module.

## 4.1 Inception feature extraction module

Fig. 2 illustrates the structure of IFE module, which is designed based on the Inception structure [18]. In HF signal detection, target signals typically exhibit a strip-distribution with fixed bandwidth and duration. However, the backbone of original YOLO employs multiple fixed-size $3 \times 3$ convolutional kernels. This single-scale, square-shaped receptive field does not align well with the actual signal distribution. It struggles to capture complete information in time domain and responds to irrelevant background noise, limiting the ability to perceive target signals. The IFE module alleviates this limitation by using parallel multi-scale branches to extract features at various scales, which enriches feature representation.

The IFE module employs multi-scale convolutional kernels ($3 \times 3$, $3 \times 5$ and $3 \times 7$) and a max-pooling branch to capture signal features across various time spans and enhance the model's ability to distinguish target signals from background noise on a global scale. The outputs from all branches are concatenated and fused via a $1 \times 1$ point-wise convolution to facilitate cross-channel interaction. To address redundancy caused by increasing channel depth, the IFE module adopts a group convolution strategy that divides input feature maps into four channel-wise groups corresponding to the multi-scale branches, thereby reducing computational cost.

In the implementation of the HF-YOLO algorithm, the IFE module replaces some CBS structures. This allows the backbone network to maintain a lightweight design while more accurately extracting signal distribution features in WTFI.

## 4.2 Time-frequency attention module

The time-frequency image integrates time and frequency information into a two-dimensional (2D) format, with horizontal axis for time, vertical axis for frequency and brightness

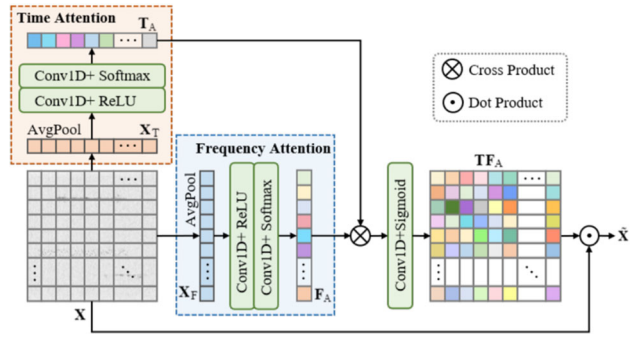



**Fig. 3** Structure of TFA module.

for signal energy intensity. Different signals and noise exhibit distinct texture features in the time-frequency domain. Therefore, effective HF signal detection requires extracting both local features and time-frequency prior information. The TFA mechanism [19] computes attention maps along time and frequency dimensions, allowing the model to focus more effectively on key time frames and frequency components of target signals. Fig. 3 illustrates the structure of the TFA module.

Assume the input feature map is $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$, where $C$ is the number of channels, $H$ and $W$ are frequency and time frame indices. Perform average pooling on $\mathbf{X}$ along the frequency dimension to get the temporal statistical features $\mathbf{X}_\mathrm{T} \in \mathbb{R}^{C \times 1 \times W}$ of each channel:

$$\mathbf{X}_\mathrm{T}(c, 1, w) = \frac{1}{H} \sum_{h=1}^{H} \mathbf{X}(c, h, w) \tag{2}$$

Then the $1 \times 1$ convolution is used to learn nonlinear inter-channel dependencies, and apply the Softmax function to map the features to a set of competitive time attention distributions $\mathbf{T}_\mathrm{A} \in \mathbb{R}^{C \times 1 \times W}$ with a sum of 1:

$$\mathbf{T}_\mathrm{A} = \mathrm{Softmax}(\mathrm{ReLU}(\mathbf{X}_\mathrm{T} \cdot \mathbf{W}_1^\mathrm{T}) \cdot \mathbf{W}_2^\mathrm{T}) \tag{3}$$

where, $\mathbf{W}_1^\mathrm{T} \in \mathbb{R}^{C \times (C/r)}$, $\mathbf{W}_2^\mathrm{T} \in \mathbb{R}^{(C/r) \times C}$, $r$ denotes the channel compression ratio. In this paper, $r = 8$.

Similarly, the generation of frequency attention distribution $\mathbf{F}_\mathrm{A} \in \mathbb{R}^{C \times H \times 1}$ can be expressed as:

$$\mathbf{X}_\mathrm{F}(c, h, 1) = \frac{1}{W} \sum_{w=1}^{W} \mathbf{X}(c, h, w) \tag{4}$$

$$\mathbf{F}_\mathrm{A} = \mathrm{Softmax}(\mathrm{ReLU}(\mathbf{X}_\mathrm{F} \cdot \mathbf{W}_1^\mathrm{F}) \cdot \mathbf{W}_2^\mathrm{F}) \tag{5}$$

After obtaining the weight distributions, matrix multiplication is used to construct fused time-frequency attention. To enhance the expressiveness of TFA, the weight distributions are fused and passed through a Sigmoid function for normalization.

The final time-frequency attention distribution $\mathbf{TF}_{\mathrm{A}} \in \mathbb{R}^{C\times H\times W}$ is given by:

$$\mathbf{TF}_{\mathrm{A}} = \sigma((\mathbf{T}_{\mathrm{A}} \otimes \mathbf{F}_{\mathrm{A}}) \cdot \mathbf{W}^{\mathrm{TF}}) \tag{6}$$

where, $\mathbf{W}^{\mathrm{TF}} \in \mathbb{R}^{C\times C}$, $\sigma$ represents the Sigmoid function. $\otimes$ denotes the element-wise tensor outer product with broadcasting mechanism. The joint attention is applied to the original feature map to enhance key time frames and frequency components, and then obtain the weighted feature map $\tilde{\mathbf{X}} \in \mathbb{R}^{C\times H\times W}$, which can be represented as:

$$\tilde{\mathbf{X}} = \mathbf{X} \cdot \mathbf{TF}_{\mathrm{A}} \tag{7}$$

In HF-YOLO, high-level features are enhanced by the TFA module and subsequently fed into the SPPF module for multi scale spatial fusion. This process guides the model to focus on regions containing target signals, improving detection robustness for signals with varying time durations and frequency distributions.

### 4.3 Loss function

After adequate feature extraction and fusion of the WTFI, HF-YOLO calculates losses for three detection heads respectively. The total loss function comprises three parts: localization loss $\mathcal{L}_{\mathrm{box}}$, classification loss $\mathcal{L}_{\mathrm{cls}}$ and confidence loss $\mathcal{L}_{\mathrm{obj}}$, which can be expressed as:

$$\mathcal{L} = \sum_{l\in\{0,1,2\}} (\lambda_{\mathrm{box}} \cdot \mathcal{L}^{l}_{\mathrm{box}} + \lambda_{\mathrm{cls}} \cdot \mathcal{L}^{l}_{\mathrm{cls}} + \lambda_{\mathrm{obj}} \cdot \mathcal{L}^{l}_{\mathrm{obj}}) \tag{8}$$

where, $\lambda_{\mathrm{box}}$, $\lambda_{\mathrm{cls}}$ and $\lambda_{\mathrm{obj}}$ are the weight coefficients for the corresponding losses. In this paper, $\lambda_{\mathrm{box}} = 7.5$, $\lambda_{\mathrm{cls}} = 0.5$ and $\lambda_{\mathrm{obj}} = 1.5$. The classification and confidence losses are computed using binary cross-entropy to distinguish target signals from background noise. The localization loss evaluates the overlap between predicted and ground-truth bounding boxes by Intersection over Union (IoU). HF-YOLO adopts the Efficient Intersection over Union (EIoU) loss [20], which enables faster and more accurate box regression by incorporating an aspect ratio optimization term.

## 5 Experiments

This section presents experiments conducted on a generated HF signal dataset to validate the proposed HF-YOLO algorithm. We compare HF-YOLO with several mainstream object detection algorithms to assess its detection accuracy in complex HF environments. In addition, ablation studies are performed to evaluate the individual contributions of the IFE and TFA modules.

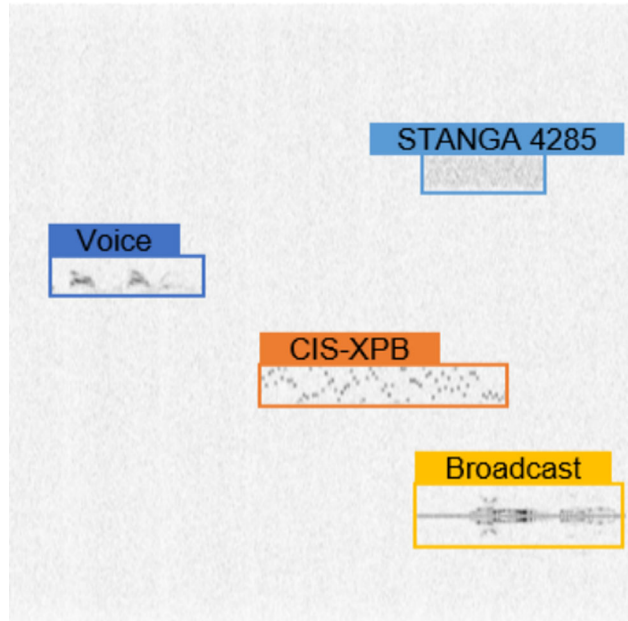



**Fig. 4** HF signal sample.

### 5.1 Dataset generation

Due to the scarcity of publicly available HF signal datasets, we construct a reliable HF time–frequency image dataset WTFID including eight HF protocol signals. The details of the protocol signals are presented in Table 1. During the simulation process, the Watterson HF channel model [27] is employed to replicate typical interference factors in HF communication.

In addition, we use IC-M803 HF receiver manufactured by Icom Inc. to capture HF wideband signals from various frequency bands. The noise data with minimal interference and no obvious signal components are selected as background signals, which provides a more accurate representation of the complex electromagnetic environment of HF channels. The above protocol signals are subsequently mixed with normalized background signals at randomly assigned carrier frequency, start and end time, and SNR. We simplified the HF channel by placing protocol signals with identical SNRs into the same background noise for statistical and experimental convenience. A short-time Fourier transform (STFT) is applied to generate grayscale time–frequency images. Fig. 4 presents a representative sample from the dataset. Specific parameters are provided in Table 2.

### 5.2 Experimental settings

#### 5.2.1 Implementation details

The experimental environment is summarized in Table 3. The dataset consists of 21,000 samples which are divided into

**Table 1** Parameters of HF signal

| HF signal | Modulation scheme | Bandwidth (kHz) | Symbol rate (Baud) |
|---|---|---|---|
| 2G-ALE [21] | 8FSK | 2.0 | 125 |
| LINK-11 CLEW [22] | $\pi/4$-DQPSK | 2.4 | 1,125 |
| STANAG 4285 [23] | MPSK(M=2,4,8) | 2.75 | 2,400 |
| CIS-75 [24] | 2FSK | 0.3 | 75 |
| CIS-XPB [25] | 16FSK | 2.8 | 65.79 |
| HFDL [26] | MPSK(M=2,4,8) | 2.4 | 1,800 |
| Broadcast | AM | 6.0 | - |
| Voice | USB, LSB | 3.0 | - |

**Table 2** Simulation parameter settings of HF dataset

| Parameter type | Parameter settings |
|---|---|
| Signal types | 2G-ALE, LINK-11 CLEW, STANAG 4285, CIS-75, CIS-XPB, HFDL, Broadcast, Voice |
| Sampling rate (kHz) | 96 |
| Sampling duration (s) | 3 |
| WTFI size | $640 \times 640$ |
| Time resolution (ms) | 2.61 |
| Frequency resolution (Hz) | 40 |
| Number of target signals per WTFI | $1 \sim 4$ |
| Center frequency (kHz) | $3 \sim 43$ |
| Burst duration (s) | $0.5 \sim 3$ |
| Watterson channel environment | ITU-HFMM [28][1] |
| SNR range (dB) | -10:2:10 |
| Sample number per SNR | 1,000 |

[1]ITU-HFMM refers to the channel model defined by International Telecommunication Union-Radio Communication Sector (ITU-R), which represents mid-latitude, moderate-condition HF channels characterized by a 1 ms delay spread and a 0.5 Hz Doppler spread.

**Table 3** Software and hardware environment configuration

| Experimental Environment | Relevant Parameters |
|---|---|
| Operating system | Ubuntu 22.04 |
| CPU | Intel(R) Xeon(R) CPU E5-2683 v3 @ 2.00 GHz |
| GPU | NVIDIA GeForce RTX 4080 SUPER (16G) |
| Signal processing software | MATLAB R2021b |
| Programming language | Python 3.9 |
| Deep learning framework | PyTorch 1.12 |

training, validation and test sets in an 8:1:1 ratio. The optimizer is stochastic gradient descent (SGD) with a learning rate of 0.005 and a batch size of 32. The maximum number of training epochs is set to 300. Additionally, early stopping is implemented to prevent overfitting.

### 5.2.2 Evaluation metrics

In object detection, mean average precision (mAP) is commonly used to evaluate overall detection performance. Considering the accuracy requirements in practical communication, we choose mAP@0.75 as the evaluation metric. This means calculating the average detection accuracy at an IoU threshold of 0.75.

In the HF signal detection, accurate localization in time and frequency domains is critical for subsequent demodulation. Therefore, the average parameter error $E_{\text{avg}}$ is used to assess the mean prediction deviation in key parameters. Its

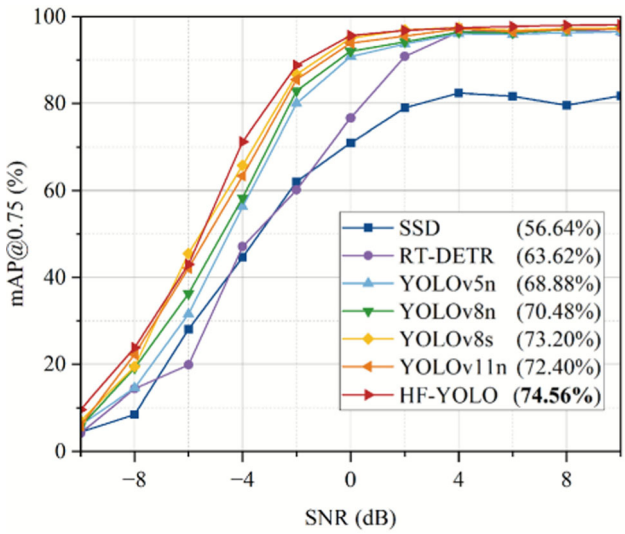



**Fig. 5** mAP@0.75 curve of various methods with SNR variations.

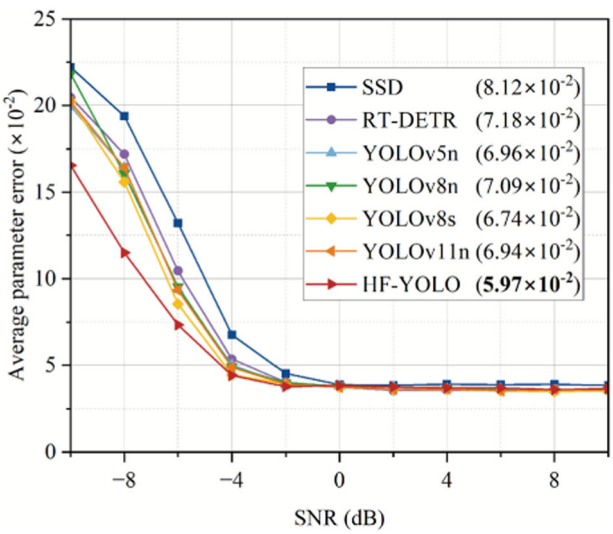



**Fig. 6** $E_{\text{avg}}$ curve of various methods with SNR variations.

calculation is shown as follows:

$$E_{\text{avg}} = \frac{1}{4\tilde{N}} \sum_{i=1}^{\tilde{N}} \left( \frac{|\tilde{f}_c^i - f_c^i|}{f_c^i} + \frac{|\tilde{B}_w^i - B_w^i|}{B_w^i} + \frac{|\tilde{t}_s^i - t_s^i|}{t_s^i} + \frac{|\tilde{t}_e^i - t_e^i|}{t_e^i} \right) \tag{9}$$

where, $\tilde{N}$ is the number of HF signals detected by the model. $f_c^i$, $B_w^i$, $t_s^i$ and $t_e^i$ represent the actual carrier center frequency, bandwidth, start time and end time of the signal, respectively. $\tilde{f}_c^i$, $\tilde{B}_w^i$, $\tilde{t}_s^i$ and $\tilde{t}_e^i$ are the corresponding predicted parameter values.

We also utilize precision and recall to assess the performance of the model. In addition, parameter count, floating point operations (FLOPs) and frames per second (FPS) are considered to evaluate its storage requirements and computational complexity.

### 5.3 Comparison with other methods

Several mainstream object detection methods are chosen as baselines, including SSD [29], RT-DETR [30], YOLOv5n [31], YOLOv8n, YOLOv8s and YOLOv11n [32]. Fig. 5 and Fig. 6 show mAP@0.75 and $E_{\text{avg}}$ variations under different SNR conditions, respectively.

As shown in Fig. 5, the detection accuracy of all algorithms improves with increasing SNR. HF-YOLO consistently achieves the highest detection accuracy, especially in low SNR ($\leq$ 0 dB), demonstrating strong anti-interference capability and robustness to weak signals. In contrast, SSD performs poorly due to its simple structure which limits its ability to extract deep features and capture contextual information. RT-DETR is sensitive to noise despite using a global attention mechanism. Its self-attention is easily distracted by noise regions in WTFI, causing significant performance degradation and severe missed detections in low SNR conditions. Fig. 6 further confirms HF-YOLO's superiority in detection accuracy. Its $E_{\text{avg}}$ remains the lowest, indicating greater precision and stability in signal parameter estimation.

Table 4 shows that HF-YOLO achieves the highest Precision and Recall, outperforming other lightweight models such as YOLOv8s and YOLOv11n. Additionally, it maintains a compact model size and fast inference speed, effectively balancing detection accuracy with real-time performance.

### 5.4 Evaluation of the proposed method

To further assess the recognition performance of the proposed HF-YOLO method across various types of HF signals, we present the confusion matrix results at SNR $= \pm 6$ dB in Fig. 7.

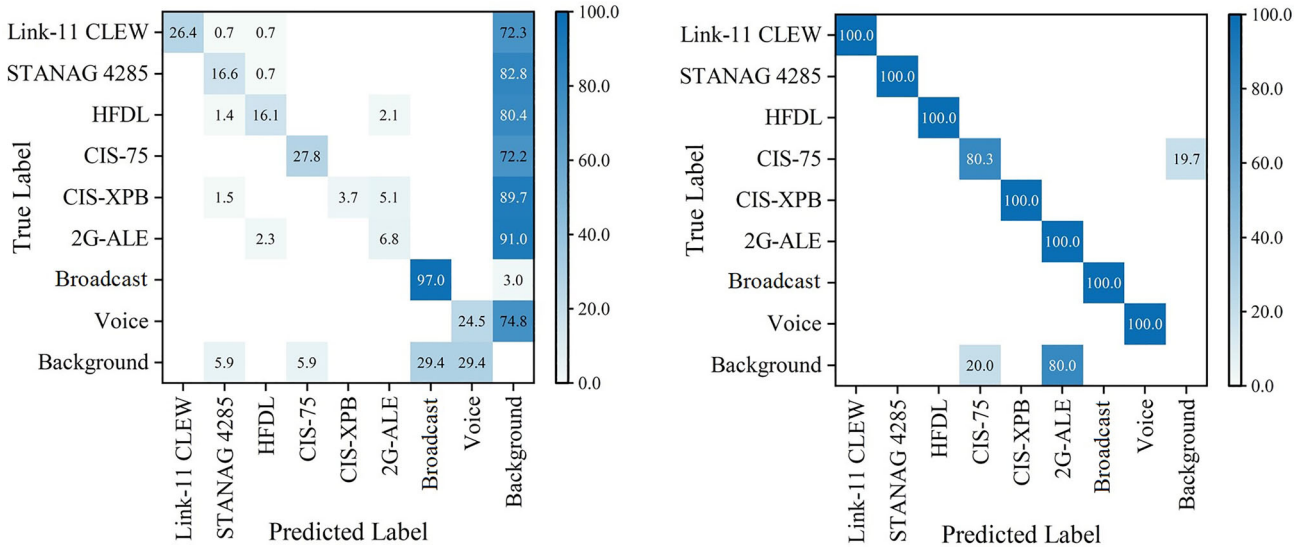
In low SNR ($-6$ dB), the model misclassifies background noise as target signals due to strong noise interference. Broadcast signals, with strong center carrier energy and distinct spectral features, maintain a high recognition rate even under these challenging conditions. In contrast, other signal types-particularly those with similar modulation characteristics, such as the CIS series, STANAG 4285 and HFDL-exhibit substantial cross-classification errors.

As the SNR increases to 6 dB, inter-signal confusion is substantially reduced. However, false alarms still occur for CIS-75 and 2G-ALE signals. This is primarily due to their frequency-hopping characteristics, which can be easily mistaken for non-stationary noise. Furthermore, the CIS-75 signal has an extremely narrow bandwidth, making it suscep-

**Table 4** Comparison of Precision, Recall and computational complexity

| Algorithm | Precision (%) | Recall (%) | Parameters (M) | FLOPs (G) | FPS (frame/s) |
|---|---|---|---|---|---|
| SSD | 77.33 | 69.33 | 24.08 | 31.38 | 25.85 |
| RT-DETR | 79.07 | 70.38 | 13.96 | 38.20 | 23.78 |
| YOLOv5n | 87.03 | 75.39 | **2.51** | 7.19 | 86.66 |
| YOLOv8n | 87.22 | 76.32 | 3.01 | 8.20 | 75.36 |
| YOLOv8s | 87.55 | 76.95 | 11.14 | 28.66 | 37.27 |
| YOLOv11n | 87.74 | 75.78 | 2.62 | **6.61** | **93.46** |
| HF-YOLO | **89.24** | **78.29** | 2.94 | 7.94 | 82.51 |

Source: This is an example of table footnote. This is an example of table footnote.




(a) SNR = −6 dB (b) SNR = 6 dB

**Fig. 7** Confusion matrix of HF-YOLO.

**Table 5** Detailed backbone structure

| Structure | Precision (%) | Recall (%) | mAP@0.75 (%) | $E_{avg}(\times 10^{-2})$ |
|---|---|---|---|---|
| [C, C, C, C] | 83.08 | **68.43** | 70.48 | 7.09 |
| [C, C, C, I] | 81.95 | 68.27 | 69.27 | 6.89 |
| [C, C, I, I] | 83.92 | 67.93 | 71.04 | 6.81 |
| [C, I, I, I] | **84.14** | 68.04 | **73.20** | **6.75** |
| [I, I, I, I] | 80.64 | 66.53 | 64.87 | 8.22 |

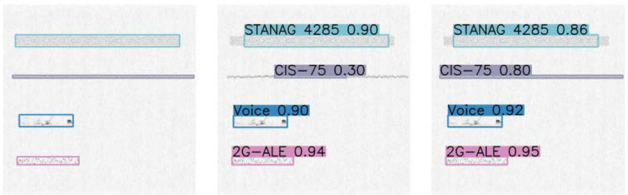
**Table 6** Number of parameters and FLOPs reduced by different backbone structures

| Structure | Parameters (K) | FLOPs (M) |
|---|---|---|
| [C, C, C, C] | - | - |
| [C, C, C, I] | -134.14 | -107.32 |
| [C, C, I, I] | -167.42 | -213.81 |
| [C, I, I, I] | -175.62 | -318.67 |
| [I, I, I, I] | -177.60 | -420.25 |

**Table 7** Performance comparison of integrating different attention mechanisms

| Network model | Precision (%) | Recall (%) | mAP@0.75 (%) | $E_{avg}(\times 10^{-2})$ |
|---|---|---|---|---|
| YOLOv8n+IFE | 84.14 | 68.04 | 73.20 | 6.75 |
| YOLOv8n+TFA | 83.67 | 68.81 | 71.66 | 6.51 |
| YOLOv8n+IFE+TA | 84.18 | 69.57 | 73.16 | 6.42 |
| YOLOv8n+IFE+FA | 83.43 | 69.71 | 72.91 | 6.20 |
| YOLOv8n+IFE+TFA | **85.53** | **70.96** | **74.56** | **5.97** |

(a) SNR = −6 dB (left: label; medium: YOLOv8n; right: HF-YOLO)




(b) SNR = 6 dB (left: label; medium: YOLOv8n; right: HF-YOLO)

**Fig. 8** Comparison of recognition results of YOLOv8n and HF-YOLO.

tible to being misclassified as narrowband interference under complex conditions.

Fig. 8 presents recognition results of YOLOv8n and HF-YOLO on the same WTFI sample. At SNR = −6 dB, target signals are nearly buried in noise, leading to low detection confidence. Nevertheless, HF-YOLO successfully detects CIS-XPB signals that are missed by YOLOv8n, demonstrating superior noise resistance. As the SNR increases to 6 dB, the time-frequency features of various signals become more distinguishable. Although both models are able to detect and recognize signals effectively, HF-YOLO provides more accurate estimation of start and end time, confirming its superiority in positioning and detection.

## 5.5 Ablation study

### 5.5.1 Layer-wise replacement of IFE module

Given that the CBS module in the first stage is responsible for preliminary feature extraction of the input signal, it is retained in the experiment to provide complete and rich original features for subsequent deep network. In stages two through five, the CBS modules are progressively replaced by the IFE module. The detection performance and computational complexity of each model are listed in Table 5 and Table 6, where, [C, C, C, C] represents the original YOLOv8n, "C" indicates the CBS module and "I" denotes the IFE module.

Results indicate that IFE modules reduce computational complexity and effectively distinguish target signals from background noise. The [C, I, I, I] structure shows the most improvement in mAP@0.75. However, completely replacing the CBS modules with IFE causes decreased accuracy due to the loss of local detail information. Considering detection performance and inference efficiency, [C, I, I, I] is chosen as the backbone for HF-YOLO.

### 5.5.2 Effectiveness of TFA module

Table 7 presents performance variations after integrating different attention mechanisms into YOLOv8n. The TFA module enhances detection probability by accurately perceiving signal duration and focusing on energy-dense frequency bands. It also improves bounding box localization and reduces parameter estimation errors. Compared to the use of IFE module alone, combining TFA with IFE preserves the advantages of multi scale feature extraction while mitigating weak signal response deficiencies.

## 6 Conclusion

This paper proposes a wideband HF signal detection method HF-YOLO. The IFE module based on group convolutions and multi scale mechanism enhances the ability to perceive time-frequency features while maintaining low computational costs. The TFA module captures feature dependencies along the time axis within a narrow frequency bandwidth and suppresses out-of-band noise. Ablation studies and comparative experiments demonstrate the superior performance of HF-YOLO. Furthermore, we constructed a specialized simulation dataset WTFID for HF signal detection. In real-world environments, target signal intensities vary and stronger signals may obscure the detection of weaker ones. To address this, we plan to improve our dataset to better reflect real-world conditions and explore methods for improving detection in the presence of strong signals that may mask weaker ones.

Moreover, various noise sources and non-target signals, coupled with the challenges of collecting large-scale HF signal data, pose significant obstacles in practical scenarios. Future research will focus on detecting HF signals in few-shot scenarios and developing strategies to mitigate interference from non-target signals.

**Author Contributions** L. G. wrote the main manuscript text and implemented the proposed HF-YOLO model, T. C. prepared figures 1–8 and curated the WTFID dataset and K. L. supervised the project and critically revised the manuscript. All authors reviewed and approved the final manuscript.

**Funding** No funding was received for this study.

**Data Availability** The wideband time-frequency image dataset (WTFID) was synthesized in-house from two sources: high frequency(HF) noise segments collected by our HF receiver and MATLAB-generated target signals. Owing to the absence of personal or third-party proprietary information, the dataset can be shared on request. All simulation scripts and anonymized data files are available from the corresponding author (liukai@shu.edu.cn) under a non-commercial, academic-use licence.

## Declarations

**Competing interests** The authors declare no competing interests.

## References

1. Wang, J., Ding, G., Wang, H.: HF communications: Past, present, and future. China Communications **15**(9), 1–9 (2018). https://doi.org/10.1109/CC.2018.8456447
2. Wang, J., Shi, Y., Yang, C., Feng, F.: A review and prospects of operational frequency selecting techniques for hf radio communication. Adv. Space Res. **69**(8), 2989–2999 (2022). https://doi.org/10.1016/j.asr.2022.01.026
3. Uysal, M., Heidarpour, M.R.: Cooperative communication techniques for future-generation HF radios. IEEE Commun. Mag. **50**(10), 56–63 (2012). https://doi.org/10.1109/MCOM.2012.6316776
4. Dearlove, C.M., Wright, P.G.: Rapid HF waveform identification. In: IEE Colloquium on Frequency Selection and Management Techniques for HF Communications, pp. 10–1105 (1999). DOI: 10.1049/ic:19990077
5. Giesbrecht, J.E.: Entropic distance for automatic modulation recognition of HF signals. In: The Institution of Engineering and Technology 11th International Conference on Ionospheric Radio Systems and Techniques (IRST 2009), pp. 1–5 (2009). DOI: 10.1049/cp.2009.0055
6. Bezruk, V., Ivanenko, S., Fedorov, O., Němec, Z., Pidanič, J.: Detection and recognition of signals in HF radio monitoring. In: 2019 29th International Conference Radioelektronika (RADIOELEKTRONIKA), pp. 1–5 (2019). DOI: 10.1109/RADIOELEK.2019.8733445
7. Li, W., Wang, K., You, L., Huang, Z.: A new deep learning framework for HF signal detection in wideband spectrogram. IEEE Signal Processing Letters 29, 1342–1346 (2022) 10.1109/LSP.2022.3179958
8. Redmon, J., Divvala, S., Girshick, R., Farhadi, A.: You Only Look Once: Unified, Real-Time Object Detection. In: 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 779–788 (2016). DOI: 10.1109/CVPR.2016.91
9. O'Shea, T., Roy, T., Clancy, T.C.: Learning robust general radio signal detection using computer vision methods. In: 2017 51st Asilomar Conference on Signals, Systems, and Computers, pp. 829–832 (2017). DOI: 10.1109/ACSSC.2017.8335463
10. Li, R., Hu, J., Li, S., Chen, S., He, P.: Blind detection of communication signals based on improved YOLO3. In: 2021 6th International Conference on Intelligent Computing and Signal Processing (ICSP), pp. 424–429 (2021). DOI: 10.1109/ICSP51882.2021.9408998
11. Zha, X., Peng, H., Qin, X., Li, G., Yang, S.: A deep learning framework for signal detection and modulation classification. Sensors 19(18) (2019) 10.3390/s19184042
12. Prasad, K.N.R.S.V., D'souza, K.B., Bhargava, V.K.: A downscaled Faster-RCNN framework for signal detection and time-frequency localization in wideband RF systems. IEEE Transactions on Wireless Communications 19(7), 4847–4862 (2020) 10.1109/TWC.2020.2987990
13. Huang, H., Li, J.-Q., Wang, J., Wang, H.: FCN-Based carrier signal detection in broadband power spectrum. IEEE Access 8, 113042–113051 (2020) 10.1109/ACCESS.2020.3003683
14. Lin, M., Zhang, X., Tian, Y., Huang, Y.: Multi-signal detection framework: A deep learning based carrier frequency and bandwidth estimation. Sensors 22(10) (2022) 10.3390/s22103909
15. Xu, Z., Savvaris, A., Tsourdos, A., Alawadi, T.: High-frequency band automatic mode recognition using deep learning. In: 2018 IEEE/AIAA 37th Digital Avionics Systems Conference (DASC), pp. 1–5 (2018). DOI: 10.1109/DASC.2018.8569367
16. Yuan, Y., Sun, Z., Wei, Z., Jia, K.: DeepMorse: A deep convolutional learning method for blind Morse signal detection in wideband wireless spectrum. IEEE Access 7, 80577–80587 (2019) 10.1109/ACCESS.2019.2923084
17. Jocher, G., Chaurasia, A., Qiu, J.: Ultralytics YOLOv8. https://github.com/ultralytics/ultralytics. <!– Missing/Wrong Year –>
18. Szegedy, C., Liu, W., Jia, Y., Sermanet, P., Reed, S., Anguelov, D., Erhan, D., Vanhoucke, V., Rabinovich, A.: Going deeper with convolutions. In: 2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 1–9 (2015). DOI: 10.1109/CVPR.2015.7298594
19. Zhang, Q., Song, Q., Ni, Z., Nicolson, A., Li, H.: Time-frequency attention for monaural speech enhancement. In: ICASSP 2022–2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 7852–7856 (2022). DOI: 10.1109/ICASSP43922.2022.9746454
20. Zhang, Y.-F., Ren, W., Zhang, Z., Jia, Z., Wang, L., Tan, T.: Focal and efficient IOU loss for accurate bounding box regression. Neurocomputing 506, 146–157 (2022) 10.1016/j.neucom.2022.07.042
21. US Department of Defense: MIL-STD-188-141A. Military Standard Interoperability and Performance Standards for Medium and High Frequency Radio Equipment (1988)
22. US Department of Defense: MIL-STD-188-203-1A. Military Standard for Tactical Digital Information Link (TADIL-A) (1988)
23. North Atlantic Treaty Organization: STANAG 4285. Characteristics of 1200/2400/3600 Bits Per Second Single Tone Modulators/Demodulators for HF Radio Links (1989)
24. diario SWL I-56578 Antonio: CIS-75 FSK 75Bd/250. https://i56578-swl.blogspot.com/search/label/CIS-75, accessed 14 August 2025. https://i56578-swl.blogspot.com/search/label/CIS-75
25. SigidWiki: CIS MFSK-16 XPB. https://www.sigidwiki.com/wiki/CIS_MFSK-16_XPB, accessed 14 August 2025. https://www.sigidwiki.com/wiki/CIS_MFSK-16_XPB
26. Airlines Electronic Engineer Committee: HF Data Link Protocols ARINC Specification 635–4. Aeronautical Radio Inc, (2003). Aeronautical Radio Inc
27. Watterson, C., Juroshek, J., Bensema, W.: Experimental Confirmation of an HF Channel Model. IEEE Trans. Commun. Tech-

nol. **18**(6), 792–803 (1970). https://doi.org/10.1109/TCOM.1970.1090438
28. International Telecommunication Union: Testing of HF Modems with Bandwidths of up to About 12 kHz Using Ionospheric Channel Simulators. (2000). International Telecommunication Union. Recommendation ITU-R F.1487
29. Liu, W., Anguelov, D., Erhan, D., Szegedy, C., Reed, S., Fu, C.-Y., Berg, A.C.: SSD: Single Shot MultiBox Detector. In: Leibe, B., Matas, J., Sebe, N., Welling, M. (eds.) Computer Vision - ECCV 2016, pp. 21–37. Springer, Cham (2016)
30. Zhao, Y., Lv, W., Xu, S., Wei, J., Wang, G., Dang, Q., Liu, Y., Chen, J.: DETRs Beat YOLOs on Real-time Object Detection. In: 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 16965–16974 (2024). DOI: https://doi.org/10.1109/CVPR52733.2024.01605
31. Jocher, G.: Ultralytics YOLOv5. DOI: 10.5281/zenodo.3908559. https://github.com/ultralytics/yolov5. <!– Missing/Wrong Year –>
32. Jocher, G., Qiu, J.: Ultralytics YOLO11. https://github.com/ultralytics/ultralytics. <!– Missing/Wrong Year –>

**Publisher's Note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.